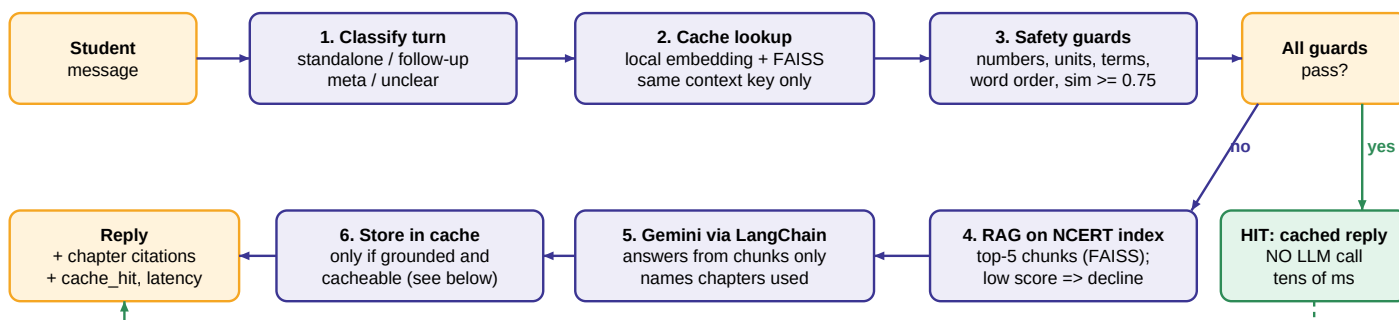


# NCERT Class 10 Science Chatbot with Smart Caching - Approach

AI Intern Round 1 (Prepzy.ai) | <Your Name> | App: <Streamlit app URL> | Code: <GitHub repo URL>



Meta turns ("explain it more simply"), unclear turns and turns after an uncached turn skip steps 2-3 and are never stored.

## How the chatbot works

The NCERT chapter PDFs are cleaned, split into ~900-character chunks tagged with their chapter, embedded with **BAAI/bge-small-en-v1.5** (local ONNX, no API cost) and stored in a **FAISS** index through LangChain. For each question the top-5 chunks are retrieved; if even the best chunk scores below a threshold the bot politely declines *without calling the LLM*. Otherwise a LangChain chain (prompt | ChatOpenAI pointed at Gemini's OpenAI-compatible endpoint | parser) answers **only** from those chunks plus recent chat history, may return an out-of-scope sentinel, and ends with a **SOURCES** line that is checked against the real chapter list to produce **citations**. **FastAPI** exposes /session and /chat; **Streamlit** shows chapter, cache-hit flag and latency for every reply and has a New conversation button.

## How the cache decides - a hit needs every check to pass

- **Turn type first.** *standalone* (self-contained), *contextual* ("what about its laws?": pronouns, "and for...", no topic), *meta* ("explain it more simply", "give an example") or *unclear* (nothing to refer to).
- **Candidate:** embedding similarity  $\geq 0.75$  against cached questions in the *same context* only.
- **Guards (deterministic):** numbers and units identical and in the same order (R = 20 vs 30 cm; u and f swapped); set of meaningful terms identical after dropping filler (what / does / mean / define), stemming and unit synonyms, so concave vs convex, NaOH vs NaCl, H<sub>2</sub>O vs H<sub>2</sub>O<sub>2</sub> all fail; term order identical when direction words appear (ice **to** water vs water **to** ice).
- **Follow-ups** are keyed to the cache entries of the previous two turns, so a follow-up is reused only after the same earlier conversation. A hit = local embedding + FAISS + guards, then return: **no LLM call**, tens of ms (target < 500 ms).

## What is cached and what is not

| What we cache                                                                                                                                                                                                                                                                             | What we never cache                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Standalone textbook answers whose best chunk scored <math>\geq 0.55</math> (well grounded).</li><li>• Follow-ups, but only under the exact preceding-turn context.</li><li>• Numericals: each distinct set of numbers is its own entry.</li></ul> | <ul style="list-style-type: none"><li>• Refusals / out-of-scope replies, clarification prompts, LLM errors.</li><li>• "Explain simpler / example / summarise": they depend on this student's last answer and a repeat needs new text.</li><li>• Any turn that follows an uncached turn (context is unknowable); answers from an older index version (re-ingest invalidates).</li></ul> |

## What did not work / trade-offs

- **Similarity threshold alone.** "Image by a concave mirror" and "...convex mirror", or R = 20 vs R = 30, differ by one token, so they embed almost identically; a threshold strict enough to block them also blocks genuine paraphrases. Hence the deterministic guards on top of embeddings.
- **Resolving follow-ups with an LLM before lookup** would turn every follow-up hit into an LLM call, breaking the "hits never call an LLM" rule. Follow-ups are keyed by conversation context instead.
- **Precision over recall.** A paraphrase that adds or swaps a content word ("refraction of light") is a miss and gets its own entry: one extra LLM call, never a wrong answer.
- **Limits:** cache is per-instance (SQLite + in-memory FAISS) and resets if the free Streamlit host is rebuilt; English only; thresholds depend on the embedding model.